SORG

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Abstract

Training-free subset selection reduces compute, yet norm-dominated contaminants can hijack geometric heuristics and trigger a catastrophic *all-outlier collapse* under norm stress and distribution shift. We present **SORG** (SoftNorm-gated Orthogonal Residual Greedy), a deterministic orthogonal-residual greedy projector that maximizes orthogonal residual gain while applying a lightweight median/MAD-based SoftNorm gate, requiring no training, gradients, or stochasticity. A projector-view analysis pinpoints the mechanism behind the norm-driven collapse and proves a *conditional clean-preference* guarantee: given a clean/outlier separation margin Δ in a robust norm statistic, the gate forms a deterministic barrier that suppresses extreme-norm points by $\exp(-p\Delta)$ while preserving projector dynamics on clean geometry. Across four flagship studies, SORG delivers large-scale vision coresets and end-to-end speedups, near-zero lethal-mode outlier selection and near-zero backdoor attack success at fixed budgets, and out-of-the-box transfer to pixels, prompts, and tokens with a broad plateau around $(\alpha, p) = (1, 2)$.

1 Introduction

Subset selection—choosing a small set of data points, pixels, tokens, or prompts—is a basic primitive for reducing compute while preserving task-relevant structure. Training-free selection is especially attractive as a plug-in, but under *norm stress* and shift, common heuristics can fail unpredictably: extreme-norm contaminants can hijack greedy pursuit and trigger a catastrophic *all-outlier collapse*, while domain shift often demands target-aware relevance rather than coverage alone.

We introduce **SORG** (SoftNorm-gated Orthogonal Residual Greedy), a deterministic OMP-like projector. SORG grows an orthonormal basis and selects the next item by maximizing *orthogonal residual progress*; its key addition is a lightweight *SoftNorm* gate based on robust statistics (median/MAD) that suppresses extreme norms, with a simple default $(\alpha, p) = (1, 2)$... TO BE FINISHED

2 Related Work

Training-free geometric coresets. Classic training-free selection targets coverage/diversity in an embedding space, e.g., farthest-first / k-center and clustering heuristics [Gonzalez, 1985, Sener and Savarese, 2018, Arthur and Vassilvitskii, 2007]. Moment-matching and herding provide deterministic summaries [Welling, 2009, Chen et al., 2010]. Submodular objectives unify many criteria [Lin and Bilmes, 2011, Wei et al., 2015], and DPPs offer probabilistic diversity [Kulesza and Taskar, 2012]. However, geometric distances can be brittle under heterogeneous norms, where contaminants dominate assignments. SORG stays training-free but replaces pure coverage with greedy *projector* progress and a robust SoftNorm gate, yielding predictable behavior under norm stress.

Greedy projector methods and data selection. SORG is closest to greedy sparse approximation, especially (O)MP [Tropp and Gilbert, 2007]. Projector-driven viewpoints also arise in column subset selection and CUR approximations [Mahoney and Drineas, 2009, Boutsidis et al., 2014], with links to volume sampling [Deshpande et al., 2006, Deshpande and Rademacher, 2010] and rank-revealing factorizations [Gu and Eisenstat, 1996]. In deep learning, subset selection is studied for training acceleration via gradient/validation matching and bilevel criteria [Mirzasoleiman et al., 2020, Killamsetty et al., 2021b,a, Jain et al., 2023]. In contrast, SORG is explicitly *training-free* and operates directly on representations, making its dynamics interpretable through projector updates and gating.

Robust and target-aware selection. Robust selection under noise, imbalance, or corruptions is often addressed via coreset/submodular objectives and validation-aware criteria [Wei et al., 2015, Mirzasoleiman et al., 2020, Killamsetty et al., 2021a]. Backdoor/poisoning results show that small high-leverage contaminants can dominate learned rules [Gu et al., 2017, Wang et al., 2019], and spectral diagnostics can expose trigger-induced directions [Tran et al., 2018]. These motivate SORG's *guided* variants: guidance aligns projector pursuit to target behavior, while SoftNorm prevents collapse onto high-norm artifacts.

Structured selection beyond dataset coresets. Selection also appears at other granularities, including token pruning/merging in transformers [Rao et al., 2021, Ryoo et al., 2021, Liang et al., 2022, Bolya et al., 2023] and ICL prompt selection [Liu et al., 2022, Rubin et al., 2022, Xu et al., 2024]. Dataset condensation synthesizes compact surrogates [Zhao and Bilen, 2021]. Across these settings, SORG contributes a single deterministic *projector-progress* primitive—selecting by residual gain rather than coverage—that remains stable under norm stress and portable across data, tokens, and prompts.

3 Algorithm – SORG: A Single Spectral Primitive for Subset Selection

SORG (*SoftNorm-gated Orthogonal Residual Greedy*) is a training-free, deterministic primitive for selecting columns from $M \in \mathbb{R}^{d\times N}$ (columns $m_1, \ldots, m_N$). Given budget k, optional guidance $r \in \mathbb{R}^d$, and optional group labels $g \in \{1, \ldots, G\}^N$, it returns $S \subseteq \{1, \ldots, N\}$ with $|S| \leq k$:

$$S = \mathrm{SORG}(k, \alpha, p)(M; r, g),$$

where $\alpha \in \mathbb{R}$ controls residual-energy removal (internally clipped to $[0,1]$) and $p \in \mathbb{R}$ controls SoftNorm gating (any $p \leq 0$ disables the gate; default $(\alpha, p) = (1, 2)$). SORG has no learned parameters and no randomness; with a fixed numerical backend the output is deterministic (ties break to the smallest index). Here "spectral" refers to orthogonal-projector/Gram geometry: SORG greedily increases the captured Frobenius energy $\|P_S M\|_F^2$ (equivalently, $\mathrm{tr}(M^\top P_S M)$), a quantity expressible via singular-value energy, even though we never compute eigenvalues or an SVD. Any feature standardization is performed upstream; SORG never rescales M.

Unguided projector objective. Let P_S be the orthogonal projector onto $\mathrm{span}(M_S)$. Unguided selection can be viewed as increasing captured energy

$$F(S; M) = \|P_S M\|_F^2 = \sum_{j=1}^{N} \|P_S m_j\|_2^2,$$

via greedy orthogonal residual updates.

Canonicalization. SORG assumes candidates are columns but applies at most one transpose to match $M \in \mathbb{R}^{d \times N}$, $r \in \mathbb{R}^d$ (if given), and $g \in \{1, \ldots, G\}^N$ (if given); see Appendix **??**.

3.1 Core Primitive (Unguided / Guided)

SORG maintains a numerically orthonormal basis $Q_t = [u_1, \ldots, u_t]$, base norms $\nu(j) = \|m_j\|_2^2$ (for gating only), and a residual cache $\gamma_t(j)$. Initialize $Q_0 = [\,]$, $S_0 = \varnothing$, and $\nu(j) = \gamma_0(j) = \|m_j\|_2^2$. Let $a \leftarrow \mathrm{clip}(\alpha, 0, 1)$ and $f \leftarrow a(2-a)$, and define

$$V_t = \{j \notin S_t : \gamma_t(j) > \varepsilon\}.$$

Stop early if $V_t = \varnothing$ (no candidates with residual energy above ε).

Unguided scoring uses $\mathrm{signal}_t(j) = \gamma_t(j)$. Guided scoring maintains a *residual* guidance vector r_t (initialized at r) and correlations $c_t(j) = m_j^\top r_t$, and uses $\mathrm{signal}_t(j) = c_t(j)^2/(\gamma_t(j) + \varepsilon)$. With SoftNorm weights $w_t(j)$ (Section 3.2), SORG selects $j_t = \arg\max_{j \in V_t} \mathrm{signal}_t(j)\, w_t(j)$. Let $u_t \propto (I - Q_t Q_t^\top) m_{j_t}$ be the stabilized residual direction and set $Q_{t+1} = [Q_t \;\; u_t]$. Writing $d_t(j) = u_t^\top m_j$, the cache update is $\gamma_{t+1}(j) = \max\{0, \gamma_t(j) - f\, d_t(j)^2\}$. When $\alpha = 1$ (hence $a = f = 1$), the cache is exact: $\gamma_t(j) = \|(I - P_{S_t}) m_j\|_2^2$ and $\gamma_{t+1}(j) = \max\{0, \gamma_t(j) - (u_t^\top m_j)^2\}$; see Appendix **??**. In guided mode, letting $s_t = u_t^\top r_t$, update $r_{t+1} = r_t - a\, u_t s_t$ and $c_{t+1}(j) = c_t(j) - a\, s_t\, d_t(j)$. We emphasize that α controls the residual-cache dynamics (and, in guided mode, the residual update of r_t), while the basis Q_t is maintained orthonormal for numerical stability. Algorithm 1 summarizes the routine.

Algorithm 1 SORG (core routine)

Require: $M \in \mathbb{R}^{d \times N}$, budget k, α, p, ε; optional (r, g)

Canonicalize shapes (at most one transpose); error on mismatch
if g is provided **then**
  Compute balanced quotas $\{q_g\}$ with $\sum_g q_g \le k$
  For each group, run the greedy loop below on $M^{(g)}$ with budget q_g using fresh local state (and the same guidance r if provided)
  return union of selected indices (mapped back to original indices)
end if
$a \leftarrow \mathrm{clip}(\alpha, 0, 1)$; $f \leftarrow a(2-a)$
$\nu(j) \leftarrow \|m_j\|_2^2$; $\gamma(j) \leftarrow \nu(j)$; $Q \leftarrow [\,]$; $S \leftarrow \varnothing$
if r is provided **then**
  $r_{\mathrm{res}} \leftarrow r$; $c(j) \leftarrow m_j^\top r_{\mathrm{res}}$ for all j
end if
for $t = 1$ to k **do**
  $V \leftarrow \{j \notin S : \gamma(j) > \varepsilon\}$; **if** $V = \varnothing$ **break**
  $w \leftarrow \mathrm{SoftNorm}(\nu, V; p)$
  if r is provided **then**
    $\mathrm{signal}(j) \leftarrow c(j)^2/(\gamma(j) + \varepsilon)$
  else
    $\mathrm{signal}(j) \leftarrow \gamma(j)$
  end if
  $j^\star \leftarrow \arg\max_{j \in V}\ \mathrm{signal}(j)\, w(j)$; $S \leftarrow S \cup \{j^\star\}$ (ties to smallest index)
  $u \leftarrow \mathrm{StabilizedNormalize}((I - QQ^\top) m_{j^\star}; \varepsilon)$;
  if degenerate **break**; $Q \leftarrow [Q\ u]$
  $d(j) \leftarrow u^\top m_j$ for all j; $\gamma(j) \leftarrow \max\{0, \gamma(j) - f\, d(j)^2\}$
  if r is provided **then**
    $s \leftarrow u^\top r_{\mathrm{res}}$; $r_{\mathrm{res}} \leftarrow r_{\mathrm{res}} - a\, u s$; $c(j) \leftarrow c(j) - a\, s\, d(j)$
  end if
end for
return S

3.2 SoftNorm Gate

SoftNorm downweights extreme norm outliers over the current valid set V using robust statistics. Let $n(j) = \sqrt{\max\{\nu(j), 0\}}$ and $\delta = 10^{-12}$. Define $b = \mathrm{median}\{n(j) : j \in V\} + \delta$, $z(j) = n(j)/b$, and $z_V = \{z(j) : j \in V\}$. Let $\mathrm{MAD}(z_V) = \mathrm{median}\{|z - \mathrm{median}(z_V)| : z \in z_V\} + \delta$ and $\theta = \mathrm{median}(z_V) + C \cdot \mathrm{MAD}(z_V)$ with fixed $C = 4$. Weights are

$$w(j) = \begin{cases} 1, & p \le 0, \\ \dfrac{1}{1 + \exp\big(p(z(j) - \theta)\big)}, & p > 0, \end{cases} \qquad \in (0, 1],$$

with numerical stabilization detailed in Appendix **??**.

3.3 Grouped Mode and Complexity

With group labels g, SORG allocates balanced quotas $\{q_g\}$ with $\sum_g q_g \le k$ (clipping by group sizes and redistributing leftovers) and runs the same core routine within each group, returning the union. Thus $|S| \le k$, with equality unless $N < k$ or early stopping occurs. Each greedy step costs one pass of dot-products $u_t^\top M$, i.e., $O(dN)$ time; hence $O(kdN)$ worst-case time. Extra memory beyond storing M is $O(dk + N)$ for the basis and cached vectors (and $O(d)$ for r in guided mode). Grouped mode preserves the same asymptotic order since $\sum_g q_g \le k$.

4 Theoretical Analysis: A Projector View of SORG

We analyze SORG (Section 3) through three pillars: (i) rank-one projector dynamics, (ii) SoftNorm as a deterministic robustness barrier, and (iii) a unified quadratic objective that explains cross-task behavior. Proofs are deferred to Appendix A.

Setup. Let $M \in \mathbb{R}^{d\times N}$ have columns $m_1, \dots, m_N$. At step t, SORG maintains an orthonormal basis $Q_t = [u_1, \dots, u_t]$ and $P_t = Q_t Q_t^\top$. Define residuals $R_t = (I - P_t)M$ and (if guided) $r_t = (I - P_t)r$. For each candidate j,

$$\gamma_t(j) := \|(I - P_t)m_j\|_2^2, \qquad c_t(j) := m_j^\top r_t. \tag{1}$$

Unguided SORG uses $\text{signal}_t(j) = \gamma_t(j)$, while guided SORG uses

$$\text{signal}_t(j) := \frac{c_t(j)^2}{\gamma_t(j) + \varepsilon}. \tag{2}$$

We denote the valid set by

$$V_t := \{j \notin S_t : \gamma_t(j) > \varepsilon\}. \tag{3}$$

4.1 Rank-One Projector Dynamics

Theorem 1 (Exact decompositions ($\alpha = 1$, gate off))**.** *Run SORG for T steps with $\alpha = 1$ and $p = 0$. Let u_{t+1} be the selected unit direction at step t. Then*

$$\begin{aligned} \|P_T M\|_F^2 &= \sum_{t=0}^{T-1} \|u_{t+1}^\top R_t\|_2^2, \\ \|P_T r\|_2^2 &= \sum_{t=0}^{T-1} (u_{t+1}^\top r_t)^2. \end{aligned} \tag{4}$$

Equivalently, $\|M\|_F^2 = \|P_T M\|_F^2 + \|R_T\|_F^2$ and $\|r\|_2^2 = \|P_T r\|_2^2 + \|r_T\|_2^2$.

Implication. Greedy projector updates admit an *exact telescoping* energy/alignment decomposition, so each selected direction contributes an explicit captured amount (Appendix A.2).

Corollary 1 (Soft removal for general α (cached update))**.** *Disable the gate ($p \le 0$) and let $0 < \alpha \le 1$. Define cached residual updates with $\widetilde{R}_0 = M$ and $\widetilde{r}_0 = r$:*

$$\widetilde{R}_{t+1} = \widetilde{R}_t - \alpha u_{t+1}\big(u_{t+1}^\top \widetilde{R}_t\big),$$

$$\widetilde{r}_{t+1} = \widetilde{r}_t - \alpha u_{t+1}\big(u_{t+1}^\top \widetilde{r}_t\big).$$

Then

$$\begin{aligned} \|\widetilde{R}_{t+1}\|_F^2 &= \|\widetilde{R}_t\|_F^2 - \rho(\alpha)\,\big\|u_{t+1}^\top \widetilde{R}_t\big\|_2^2, \\ \rho(\alpha) &:= \alpha(2-\alpha) \in (0,1]. \end{aligned} \tag{5}$$

and similarly $\|\widetilde{r}_{t+1}\|_2^2 = \|\widetilde{r}_t\|_2^2 - \rho(\alpha)\big(u_{t+1}^\top \widetilde{r}_t\big)^2$.

Implication. The implementation's cached energy update corresponds to a *soft rank-one removal* with decay rate $\rho(\alpha)$, so α controls energy removal smoothly without changing the projector viewpoint (Appendix A.2).

Guided gain interpretation. For $\gamma_t(j) > 0$, letting $u_j = (I - P_t)m_j/\sqrt{\gamma_t(j)}$ gives

$$(u_j^\top r_t)^2 = \frac{c_t(j)^2}{\gamma_t(j)},$$

and (2) is its ε-stabilized proxy.

4.2 Robustness via SoftNorm

Let $v_j = \|m_j\|_2$ and define $\mathrm{base}_t := \mathrm{median}(v(V_t)) + 10^{-12}$ and $z_t(j) := v_j/\mathrm{base}_t$. Here $\mathrm{MAD}(x) := \mathrm{median}(|x - \mathrm{median}(x)|)$. Compute a robust threshold on $z_t(V_t)$ via $\mathrm{thr}_t := \mathrm{median}(z_t(V_t)) + C\,\mathrm{MAD}(z_t(V_t))$ with $C = 4$ (and 10^{-12} stabilizers as needed). The clipped SoftNorm weight is

$$\begin{aligned} w_t(j) &= \begin{cases} 1, & p \le 0, \\ \big(1 + \exp(p(z_t(j) - \mathrm{thr}_t))\big)^{-1}, & p > 0, \end{cases} \\ w_t(j) &\leftarrow \mathrm{clip}\big(w_t(j), \rho_{\min}, 1\big), \qquad j \in V_t. \end{aligned} \tag{6}$$

Theorem 2 (Clean preference (conditional))**.** *Fix step t and partition $V_t = C_t \cup O_t$ (clean vs. outlier). Assume $\Delta > 0$ such that $\max_{i \in C_t} z_t(i) \le \mathrm{thr}_t$ and $\min_{k \in O_t} z_t(k) \ge \mathrm{thr}_t + \Delta$. Let $\psi_C = \max_{i \in C_t} \mathrm{signal}_t(i)$ and $\psi_O = \max_{k \in O_t} \mathrm{signal}_t(k)$. If*

$$\psi_O \cdot \max\{\rho_{\min}, e^{-p\Delta}\} < \tfrac{1}{2}\,\psi_C, \tag{7}$$

then $\arg\max_{j \in V_t} \mathrm{signal}_t(j) w_t(j) \in C_t$.

Implication. Under clean/outlier separation in the robust norm statistic, SoftNorm enforces a *deterministic barrier*: even large outlier signals cannot win once the gate suppresses them exponentially in $p\Delta$ (Appendix A.3).

4.3 A Unified Quadratic Objective

Let $Y \in \mathbb{R}^{d \times L}$ encode task representations. For $\lambda \ge 0$, define

$$\begin{aligned} \mathcal{F}_\lambda(S) &:= \|P_S M\|_F^2 + \lambda \|P_S Y\|_F^2 \\ &= \big\|P_S[\,M \;\; \sqrt{\lambda} Y\,]\big\|_F^2. \end{aligned} \tag{8}$$

Proposition 1 (Greedy gain and guided limit)**.** *With $\alpha = 1$ and $p = 0$, the marginal gain of adding a unit direction u at step t is*

$$\Delta\mathcal{F}_\lambda = \|u^\top R_t\|_2^2 + \lambda\|u^\top Y_t\|_2^2, \qquad Y_t = (I - P_t)Y. \tag{9}$$

For $L = 1$ and $Y = r$, the greedy step is dominated by the alignment term as $\lambda \to \infty$, recovering guided SORG's criterion (up to tie-breaking and ε).

Implication. A single projector objective $\mathcal{F}_\lambda$ unifies coreset-like energy capture and task-driven alignment; guided SORG appears as an extreme-λ limit (Appendix A.4).

4.4 Hyperparameter Robustness

Lemma 1 (SoftNorm is Lipschitz in the sharpness p)**.** *Fix $z(j)$ and* thr*. For $p \geq 0$, let $w(j;p) = \big(1 + \exp(p(z(j) - \mathrm{thr}))\big)^{-1}$ and then clip it to $[\rho_{\min}, 1]$. Then for any $p_1, p_2 \geq 0$ and any j,*

$$|w(j;p_1) - w(j;p_2)| \leq \frac{|z(j) - \mathrm{thr}|}{4}\,|p_1 - p_2|.$$

Implication. Changing the gate sharpness by Δp perturbs every SoftNorm weight by at most $O(\Delta p)$, so the gate does not introduce brittle discontinuities (Appendix A.5).

Proposition 2 (Local path stability under a margin)**.** *Fix (α_0, p_0) and let $S_T(\alpha_0, p_0)$ be the subset produced by SORG in T steps. Assume the reference run has a uniform score margin $\Delta > 0$ up to step T and that the per-step score is locally Lipschitz in (α, p) at fixed state. Then there exists a neighborhood of (α_0, p_0) in which the greedy selection path is invariant: $S_t(\alpha, p) = S_t(\alpha_0, p_0)$ for all $t \leq T$.*

Implication. Hyperparameters are *path-stable*: once the best candidate is separated by a margin, small perturbations in (α, p) cannot change any selected index along the run (Appendix A.5).

5 Conclusion

We presented **SORG**, a deterministic, training-free selection primitive built around an OMP-like projector with a robust SoftNorm gate. Our projector-view analysis explains the residual-gain dynamics and the gate-induced *clean preference* that prevents norm-driven all-outlier collapse. Across four flagships, SORG yields strong vision coresets and practical speedups, remains stable under lethal contamination and backdoor regimes, and transfers to structured selection (pixels, prompts, tokens) with a broad hyperparameter plateau near $(\alpha, p) = (1, 2)$. Overall, SORG is a simple and reliable spectral primitive whenever stable "select-by-residual" behavior is required.

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A Proofs

A.1 Setup and Notation

Let $M \in \mathbb{R}^{d\times N}$ have columns $m_1, \dots, m_N \in \mathbb{R}^d$. At step t, SORG maintains an orthonormal basis $Q_t = [u_1, \dots, u_t]$ and projector $P_t = Q_t Q_t^\top$. Define residuals

$$R_t := (I - P_t)M, \qquad r_t := (I - P_t)r \text{ (if guided)}.$$

For each candidate j, recall

$$\gamma_t(j) := \|(I - P_t)m_j\|_2^2, \qquad c_t(j) := m_j^\top r_t,$$

and the valid set

$$V_t := \{j \notin S_t : \gamma_t(j) > \varepsilon\}.$$

SoftNorm weights satisfy $w_t(j) \in [\rho_{\min}, 1]$ after clipping. For hyperparameter stability statements, we assume bounded inputs $\max_j \|m_j\|_2 \le B$, $\|r\|_2 \le B_r$, and (when used) $\|Y\|_F \le B_Y$.

A.2 Rank-One Projector Dynamics

Theorem 1 (Exact decompositions; $\alpha = 1$, gate off)

Run SORG for T steps with $\alpha = 1$ and $p = 0$. Let u_{t+1} be the selected unit direction at step t. Then

$$\|P_T M\|_F^2 = \sum_{t=0}^{T-1} \|u_{t+1}^\top R_t\|_2^2, \qquad \|P_T r\|_2^2 = \sum_{t=0}^{T-1} (u_{t+1}^\top r_t)^2.$$

Equivalently, $\|M\|_F^2 = \|P_T M\|_F^2 + \|R_T\|_F^2$ and $\|r\|_2^2 = \|P_T r\|_2^2 + \|r_T\|_2^2$.

Proof of Theorem 1. With $\alpha = 1$ and $p = 0$, each step appends a direction u_{t+1} orthonormal to $\mathrm{span}(Q_t)$. Hence

$$P_{t+1} = P_t + u_{t+1}u_{t+1}^\top, \qquad R_{t+1} = (I - P_{t+1})M = R_t - u_{t+1}u_{t+1}^\top R_t.$$

By orthogonality, $u_{t+1}u_{t+1}^\top R_t$ is orthogonal to R_{t+1}, so

$$\|R_t\|_F^2 = \|u_{t+1}u_{t+1}^\top R_t\|_F^2 + \|R_{t+1}\|_F^2 = \|u_{t+1}^\top R_t\|_2^2 + \|R_{t+1}\|_F^2.$$

Telescoping from $t = 0$ to $T - 1$ yields

$$\|R_0\|_F^2 - \|R_T\|_F^2 = \sum_{t=0}^{T-1} \|u_{t+1}^\top R_t\|_2^2.$$

Since $R_0 = M$ and P_T is an orthogonal projector,

$$\|M\|_F^2 = \|P_T M\|_F^2 + \|(I - P_T)M\|_F^2 = \|P_T M\|_F^2 + \|R_T\|_F^2,$$

which implies the first identity. The second identity follows identically by replacing M with r and $\|\cdot\|_F$ with $\|\cdot\|_2$. □

Corollary 1 (Soft removal; cached update).

Disable the gate ($p \le 0$) and let $0 < \alpha \le 1$. Define cached residual updates with $\widetilde{R}_0 = M$ and $\widetilde{r}_0 = r$:

$$\widetilde{R}_{t+1} = \widetilde{R}_t - \alpha u_{t+1}\big(u_{t+1}^\top \widetilde{R}_t\big), \qquad \widetilde{r}_{t+1} = \widetilde{r}_t - \alpha u_{t+1}\big(u_{t+1}^\top \widetilde{r}_t\big).$$

Then

$$\|\widetilde{R}_{t+1}\|_F^2 = \|\widetilde{R}_t\|_F^2 - \rho(\alpha)\|u_{t+1}^\top \widetilde{R}_t\|_2^2, \quad \rho(\alpha) = \alpha(2-\alpha) \in (0,1],$$

and similarly $\|\widetilde{r}_{t+1}\|_2^2 = \|\widetilde{r}_t\|_2^2 - \rho(\alpha)\big(u_{t+1}^\top \widetilde{r}_t\big)^2$.

Proof of Corollary 1. Write the update as

$$\widetilde{R}_{t+1} = \widetilde{R}_t - \alpha u u^\top \widetilde{R}_t, \quad u := u_{t+1}.$$

Expanding the Frobenius norm and using $\|u\|_2 = 1$,

$$\begin{aligned}
\|\widetilde{R}_{t+1}\|_F^2 &= \|\widetilde{R}_t\|_F^2 - 2\alpha\langle \widetilde{R}_t, uu^\top \widetilde{R}_t\rangle + \alpha^2\|uu^\top \widetilde{R}_t\|_F^2 \\
&= \|\widetilde{R}_t\|_F^2 - 2\alpha\|u^\top \widetilde{R}_t\|_2^2 + \alpha^2\|u^\top \widetilde{R}_t\|_2^2 \\
&= \|\widetilde{R}_t\|_F^2 - \alpha(2-\alpha)\|u^\top \widetilde{R}_t\|_2^2.
\end{aligned}$$

The vector case is identical with $\widetilde{r}_t$ in place of $\widetilde{R}_t$. □

A.3 Robustness via SoftNorm

Theorem 2 (Clean preference; conditional).

Fix step t and partition $V_t = C_t \cup O_t$ (clean vs. outlier). Assume $\Delta > 0$ such that $\max_{i \in C_t} z_t(i) \le \mathrm{thr}_t$ and $\min_{k \in O_t} z_t(k) \ge \mathrm{thr}_t + \Delta$. Let $\psi_C = \max_{i \in C_t} \mathrm{signal}_t(i)$ and $\psi_O = \max_{k \in O_t} \mathrm{signal}_t(k)$. If

$$\psi_O \cdot \max\{\rho_{\min}, e^{-p\Delta}\} < \tfrac{1}{2}\,\psi_C,$$

then $\arg\max_{j \in V_t} \mathrm{signal}_t(j) w_t(j) \in C_t$; hence SORG cannot select an outlier at step t.

Proof of Theorem 2. By SoftNorm definition (before clipping), for $p > 0$,

$$w_t(j) = \frac{1}{1 + \exp(p(z_t(j) - \mathrm{thr}_t))}.$$

For any clean $i \in C_t$ we have $z_t(i) \le \mathrm{thr}_t$, hence $\exp(p(z_t(i) - \mathrm{thr}_t)) \le 1$ and therefore

$$w_t(i) \ge \frac{1}{1+1} = \frac{1}{2}.$$

For any outlier $k \in O_t$ we have $z_t(k) \ge \mathrm{thr}_t + \Delta$, hence

$$w_t(k) = \frac{1}{1 + \exp(p(z_t(k) - \mathrm{thr}_t))} \le \frac{1}{1 + \exp(p\Delta)} \le \exp(-p\Delta).$$

After clipping to $[\rho_{\min}, 1]$, we still have

$$w_t(i) \geq \tfrac{1}{2} \quad \text{and} \quad w_t(k) \leq \max\{\rho_{\min}, e^{-p\Delta}\}.$$

Let $i^\star \in \arg\max_{i \in C_t} \text{signal}_t(i)$ and $k^\star \in \arg\max_{k \in O_t} \text{signal}_t(k)$, so $\text{signal}_t(i^\star) = \psi_C$ and $\text{signal}_t(k^\star) = \psi_O$. Then the best clean score satisfies

$$\max_{i \in C_t} \text{signal}_t(i) w_t(i) \;\geq\; \psi_C \cdot \tfrac{1}{2},$$

while the best outlier score satisfies

$$\max_{k \in O_t} \text{signal}_t(k) w_t(k) \;\leq\; \psi_O \cdot \max\{\rho_{\min}, e^{-p\Delta}\}.$$

Under the stated condition, the best clean score strictly exceeds the best outlier score, so the global maximizer over V_t lies in C_t. □

A.4 Unified Quadratic Objective

Proposition 1 (Greedy gain and guided limit).

With $\alpha = 1$ and $p = 0$, the marginal gain of adding a unit direction u at step t is

$$\Delta\mathcal{F}_\lambda = \|u^\top R_t\|_2^2 + \lambda \|u^\top Y_t\|_2^2, \qquad Y_t = (I - P_t)Y.$$

For $L = 1$ and $Y = r$, the greedy step is dominated by the alignment term as $\lambda \to \infty$, recovering guided SORG's criterion (up to tie-breaking and ε).

Proof of Proposition 1. Let $P_{t+1} = P_t + uu^\top$ with $u \perp \text{range}(P_t)$ and $\|u\|_2 = 1$. Then

$$\|P_{t+1}M\|_F^2 - \|P_t M\|_F^2 = \|u^\top (I - P_t)M\|_2^2 = \|u^\top R_t\|_2^2,$$

where orthogonality removes cross terms. Similarly,

$$\|P_{t+1}Y\|_F^2 - \|P_t Y\|_F^2 = \|u^\top (I - P_t)Y\|_2^2 = \|u^\top Y_t\|_2^2.$$

Therefore

$$\Delta\mathcal{F}_\lambda = \|u^\top R_t\|_2^2 + \lambda \|u^\top Y_t\|_2^2.$$

For $L = 1$ and $Y = r$, as $\lambda \to \infty$ the gain is dominated by $\|u^\top r_t\|_2^2$. For a candidate column m_j, the new direction in guided SORG is proportional to $u_j \propto (I - P_t)m_j$. If $\gamma_t(j) = \|(I - P_t)m_j\|_2^2 > 0$, then

$$(u_j^\top r_t)^2 = \frac{\big((I - P_t)m_j\big)^\top r_t}{\|(I - P_t)m_j\|_2}^2 = \frac{c_t(j)^2}{\gamma_t(j)},$$

and guided SORG uses the stabilized proxy $c_t(j)^2/(\gamma_t(j) + \varepsilon)$, matching the $\lambda \to \infty$ alignment-dominated selection up to ε and tie-breaking. □

A.5 Hyperparameter Robustness (Stability Proofs)

Lemma 1 (Lipschitz in p).

Fix $z(j)$ and thr. Let $w(j;p) = 1/\big(1+\exp(p(z(j)-\mathrm{thr}))\big)$ for $p \geq 0$ (before clipping), and define $B_z = \max_j |z(j) - \mathrm{thr}|$. Then for any $p_1, p_2 \geq 0$ and any index j,

$$|w(j;p_1) - w(j;p_2)| \leq \frac{|z(j)-\mathrm{thr}|}{4}|p_1 - p_2|, \qquad \max_j |w(j;p_1) - w(j;p_2)| \leq \frac{B_z}{4}|p_1 - p_2|.$$

Both inequalities remain valid after clipping $w(j;p)$ to $[\rho_{\min}, 1]$.

Proof of Lemma 1. Consider $g(t) = 1/(1+e^t)$, so $w(j;p) = g(p(z(j)-\mathrm{thr}))$. We have

$$g'(t) = -\frac{e^t}{(1+e^t)^2}, \qquad |g'(t)| \leq \frac{1}{4} \text{ for all } t \in \mathbb{R}.$$

By the chain rule,

$$\left|\frac{\partial}{\partial p} w(j;p)\right| = |g'(p(z(j)-\mathrm{thr}))| \cdot |z(j)-\mathrm{thr}| \leq \frac{|z(j)-\mathrm{thr}|}{4}.$$

Applying the mean value theorem gives

$$|w(j;p_1) - w(j;p_2)| \leq \frac{|z(j)-\mathrm{thr}|}{4}|p_1 - p_2|.$$

Taking the maximum over j yields the second inequality. Finally, scalar clipping is 1-Lipschitz, so the bounds remain valid after clipping. □

Proposition 2 (Local path stability under a margin).

Fix (α_0, p_0) and let $S_T(\alpha_0, p_0)$ be the subset produced by SORG in T steps. Assume the reference run has a uniform margin $\Delta > 0$ up to step T and that the per-step score is locally Lipschitz in (α, p) at fixed state. Then there exists a neighborhood of (α_0, p_0) in which the greedy selection path is invariant: $S_t(\alpha, p) = S_t(\alpha_0, p_0)$ for all $t \leq T$.

Proof of Proposition 2. Let $\mathrm{score}_t(j;\alpha,p)$ denote the step-t score assigned to candidate j by SORG when evaluated on a *fixed* step-t state (i.e., with Q_t, caches, V_t, and guided residuals held fixed). By the local Lipschitz assumption, for each $t \leq T$ there exists $L_t < \infty$ and a neighborhood of (α_0, p_0) such that

$$\max_{j \in V_t} \big|\mathrm{score}_t(j;\alpha,p) - \mathrm{score}_t(j;\alpha_0,p_0)\big| \leq L_t\big(|\alpha-\alpha_0| + |p-p_0|\big).$$

Let $j_t^\star$ be the (unique) maximizer under (α_0, p_0) at step t, and assume a uniform margin $\Delta > 0$ along the reference run:

$$\mathrm{score}_t(j_t^\star;\alpha_0,p_0) \;\geq\; \max_{j \in V_t,\ j \neq j_t^\star} \mathrm{score}_t(j;\alpha_0,p_0) + \Delta \qquad \text{for all } t \leq T.$$

Choose (α, p) so that $|\alpha-\alpha_0| + |p-p_0| \leq \Delta/(4\max_{t\leq T} L_t)$. Then for each $t \leq T$ we have

$$\mathrm{score}_t(j_t^\star;\alpha,p) \geq \mathrm{score}_t(j_t^\star;\alpha_0,p_0) - \Delta/4, \qquad \mathrm{score}_t(j;\alpha,p) \leq \mathrm{score}_t(j;\alpha_0,p_0) + \Delta/4 \;\; (j \neq j_t^\star),$$

hence the maximizer at step t remains $j_t^\star$.

It remains to justify that the "fixed state" used above is indeed the same across runs up to step T. Since SORG's update is deterministic given the chosen index (and a fixed numerical backend), if the selected index at step t matches the reference ($j_t = j_t^\star$), then the entire internal state used to evaluate step $t+1$ (including Q_t, cached γ, the valid set V_{t+1}, and guided residual variables if present) also matches the reference. Because the argument above shows that $j_t = j_t^\star$ for every $t \leq T$, it follows inductively that the states coincide at each step, and therefore $S_t(\alpha, p) = S_t(\alpha_0, p_0)$ for all $t \leq T$.

□